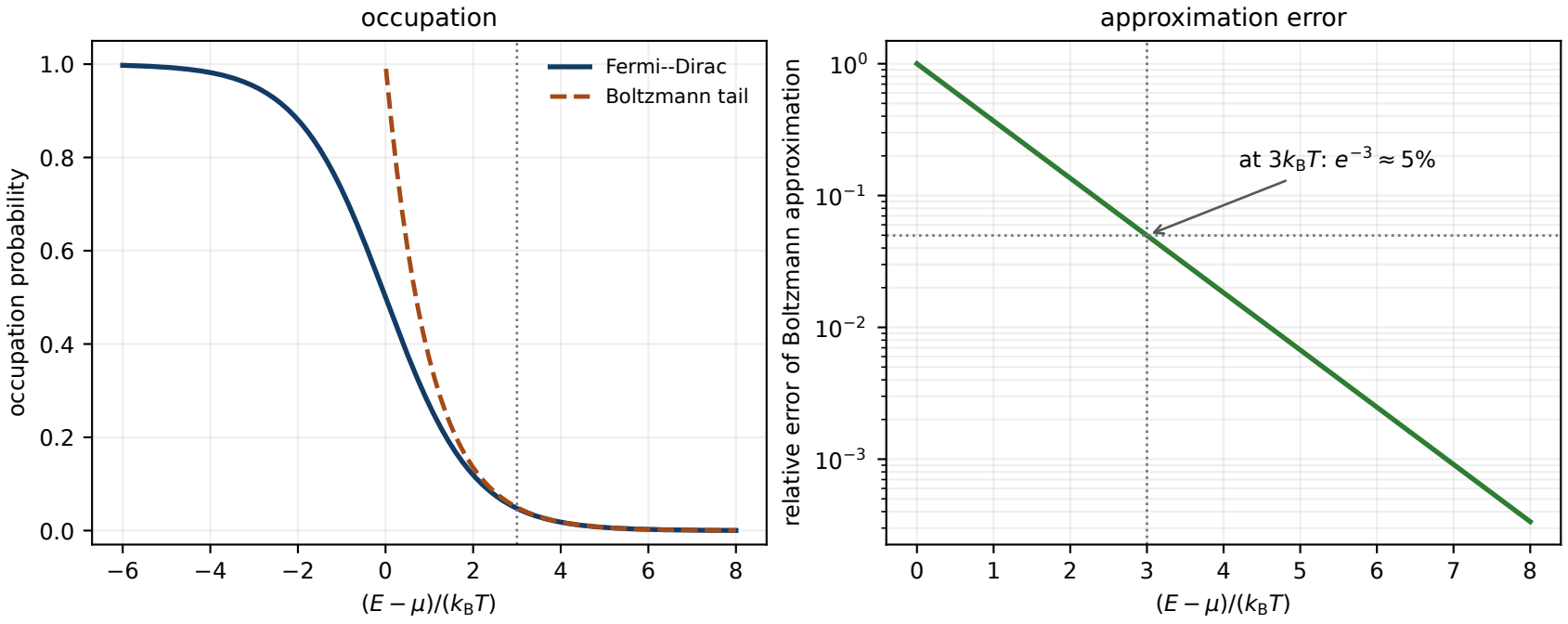
occupation
occupation probability
1.0
0.8
0.6
0.4
0.2
0.0
−6
−4
−2
0
2
4
6
8
$(E-\mu)/(k_\mathrm{B}T)$
Fermi--Dirac
Boltzmann tail
approximation error
relative error of Boltzmann approximation
10^{0}
10^{-1}
10^{-2}
10^{-3}
at $3k_\mathrm{B}T$: $e^{-3} \approx 5\%$
0
1
2
3
4
5
6
7
8
$(E-\mu)/(k_\mathrm{B}T)$